TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Size Percentage in Fabric

How much of the woven cloth is warp size waiting to be removed: identification, enzymatic desizing, and the two reporting conventions that must never be confused.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 4 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to identify the size present on a woven cotton fabric, to remove it completely by desizing and washing, and to determine the size percentage from dry weights before and after removal, expressed on both the desized-weight and sized-weight bases.

1.1 Why size is there — and why it must come out

Warp yarns are sized before weaving: a film of size — traditionally starch, often blended or replaced by polyvinyl alcohol (PVA), carboxymethyl cellulose (CMC) or acrylic copolymers, with lubricants and humectants — glues down protruding fibres, raises abrasion resistance and lets the warp survive thousands of loom cycles. Size is a weaving aid, not a fabric component. Left in the cloth it blocks absorbency, resists dye and print paste, stiffens handle and inflates the fabric weight. The grey fabric of Experiments 1–3 carries its size; wet processing begins by removing it (**desizing**), and the size percentage tells the processor how much material must come out, how much effluent load desizing will create, and how much of the purchased grey weight was never fibre at all.

1.2 Identification of the size

The desizing method depends on the size type, so identification comes first. The spot test for **starch** is iodine: a drop of iodine–potassium iodide solution on the fabric (or on an extract) gives an intense **blue to blue-black colour** with starch. PVA gives a different response (a blue-green to reddish colour with iodine–boric acid reagent) and, practically, PVA is not removed by enzymes at all — it must be washed out with hot water and detergent, sometimes with oxidative help. CMC is water-soluble and washes out readily. A negative iodine test on a sized-feeling fabric therefore redirects the whole procedure: enzyme desizing of a PVA-sized fabric removes almost nothing and reports a false low size percentage.

1.3 Desizing

For starch size, the standard laboratory route is **enzymatic desizing**: bacterial alpha-amylase hydrolyses the starch chains into soluble dextrins and sugars that rinse away. The specimen is wetted out, treated with the enzyme bath at its working temperature (typically 60–70 °C, near-neutral to mildly acidic pH per the enzyme supplier) for long enough to degrade the film completely — confirmed by a negative iodine test on the treated fabric — then washed thoroughly in hot and cold water to remove all degradation products and loosened matter. Oxidative desizing (persulphate or peroxide based) is the mill alternative; it attacks starch and PVA alike but risks fibre damage if uncontrolled, which would corrupt a gravimetric result by removing cotton as well as size.

1.4 The gravimetric calculation — two conventions

Both weighings are of **oven-dry** fabric, so moisture never enters the difference:

$$\text{Size (g)} = \text{dry sized mass} - \text{dry desized mass}$$

$$\text{Size \% on desized weight} = \text{size} / \text{dry desized mass} \times 100 \text{ (the add-on basis)}$$

$$\text{Size \% on sized weight} = \text{size} / \text{dry sized mass} \times 100 \text{ (the content basis)}$$

The two percentages describe the same grams and are both in commercial use — add-on is the sizer's language (how much size was put on the yarn), content is the analyst's (what fraction of the cloth in hand is size). They differ by more than a percentage point at typical levels, so a size figure without its basis is an incomplete result, exactly as a thickness without its pressure is incomplete (Experiment 3).

Conditioning note: specimens are handled and identified in the standard atmosphere (65% RH, 20 °C), but the reported masses are oven-dry masses — dried to constant weight and weighed promptly from the desiccator — because regain differences between the sized and desized states would otherwise masquerade as size.

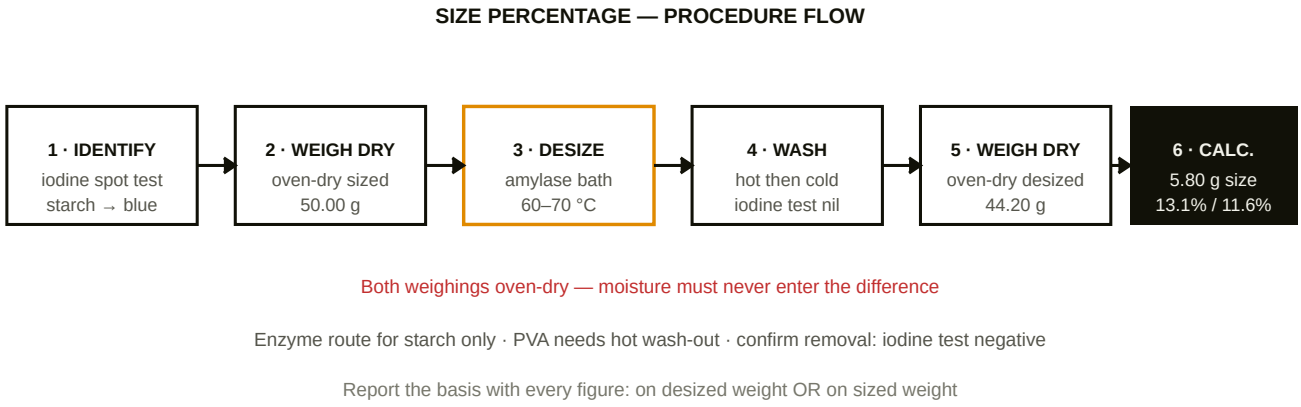


Figure 1 — The gravimetric size determination as a flow: identify the size, weigh dry, desize, wash until the iodine test is negative, weigh dry again, and calculate on a stated basis.

2. Apparatus and procedure

- **Apparatus:** balance reading to 0.01 g; drying oven and desiccator; thermostatically controlled desizing bath; iodine–potassium iodide spot-test reagent; amylase desizing agent; detergent; beakers, thermometer and conditioned fabric specimens cut clear of selvedges.
- **Identification:** place a drop of iodine reagent on the specimen. A blue to blue-black colour confirms starch size. Record the identification before choosing the desizing route.
- **First weighing:** dry the sized specimen in the oven to constant weight, cool in the desiccator and weigh promptly: dry sized mass.
- **Desizing:** wet out the specimen, immerse in the amylase bath at 60–70 °C at the supplier's dose and pH, and hold with gentle agitation until a fresh iodine spot test on the fabric is negative (commonly 30–60 min in the laboratory).
- **Washing:** wash thoroughly — hot wash with a little detergent, then repeated cold rinses — until the rinse water runs clear and no loosened size remains in the fabric.
- **Second weighing:** dry the desized specimen to constant weight exactly as before, cool in the desiccator and weigh: dry desized mass. Compute size mass and both percentage conventions.

3. Worked calculation

The iodine spot test gave an immediate deep blue — starch size — so the enzymatic route applies. Oven-dry weighings:

WEIGHING	DRY MASS
Sized specimen (before desizing)	50.00 g
Desized specimen (after desize + wash)	44.20 g
Size removed	50.00 − 44.20 = 5.80 g

$$\text{Size \% on desized weight} = 5.80 / 44.20 \times 100 = 13.1\%$$

$$\text{Size \% on sized weight} = 5.80 / 50.00 \times 100 = 11.6\%$$

Both statements are true of the same specimen: the sizer applied size equal to 13.1% of the bare yarn weight, and 11.6% of the grey cloth as purchased is size that wet processing must remove. Note the check that ties them: 44.20 × 1.131 ≈ 50.00 g — the two bases are one measurement, two fractions.

> **INTERPRETING 13.1% ADD-ON**
>
> A size add-on of about 13% on warp weight is a normal working level for a medium-count cotton warp of the kind in the reference fabric (70 EPI × 59 PPI, 185.5 g/m² in its sized grey state). Much

> lower risks loom breaks; much higher wastes size, stiffens the cloth and loads the desizing effluent. Because size sits almost entirely on the warp, the percentage on total fabric weight (11.6%) is lower than the add-on quoted at the size box — always ask which basis a figure uses.

4. Precautions

- Identify before desizing: enzyme on PVA size removes almost nothing and produces a confidently wrong low result. The iodine test takes ten seconds.
- Both weighings must be oven-dry to constant weight and taken from the desiccator; a desized fabric re-absorbs moisture faster than a sized one, and the regain difference would be counted as size.
- Continue desizing and washing until the iodine test on the fabric is negative — residual degraded starch left in the cloth is weighed as fibre and understates the size.
- Handle the wet desized specimen gently: it has lost its protective film, and fibre lost in handling is indistinguishable from size in the final difference.
- Cut specimens from the body of the fabric, clear of selvedges, where size distribution is representative; selvedge warps are often sized differently.
- Avoid harsh oxidative conditions in a gravimetric determination — fibre damage removes cotton along with size and overstates the percentage.

Practice (viva preparation)

1. How is starch size identified, and what colour is a positive test?
2. A dry sized specimen weighs 50.00 g and its dry desized mass is 44.20 g. Find the size in grams and both percentage conventions.
3. Why are the two percentages in Question 2 different when the grams of size are the same?
4. Why would an amylase bath fail on a PVA-sized fabric, and what route removes PVA?
5. Why are both weighings made oven-dry rather than at standard-atmosphere regain?
6. How is complete removal of size confirmed during the procedure?
7. What happens in dyeing and finishing if desizing is incomplete?

Answer key

1. By the iodine spot test: a drop of iodine–potassium iodide solution on the fabric or its extract turns intense blue to blue-black in the presence of starch. It is the standard first test because the desizing method is chosen from the size type it reveals.
2. Size = 50.00 − 44.20 = 5.80 g. On desized weight: 5.80/44.20 × 100 = 13.1%. On sized weight: 5.80/50.00 × 100 = 11.6%.

3. The numerator is identical (5.80 g) but the denominators differ: the desized basis divides by the bare fabric (44.20 g), the sized basis by the whole cloth including the size itself (50.00 g). Dividing by the smaller mass gives the larger percentage. Neither is wrong — but a figure quoted without its basis is ambiguous by 1.5 percentage points here.
4. Amylase is specific: it hydrolyses the alpha-glucosidic bonds of starch and has no action on the synthetic PVA chain, so the film survives the bath almost intact. PVA is removed by washing out — hot water with detergent, exploiting its water solubility — sometimes assisted by oxidative desizing agents. This is why identification precedes desizing.
5. Sized and desized cotton hold different amounts of moisture at the same atmosphere — the size film itself is hygroscopic and its removal changes regain. At 65% RH those regain differences are grams-scale on a 50 g specimen and would be booked as size. Oven-dry weighing to constant weight removes moisture from the comparison entirely; only the solid difference remains.
6. By repeating the iodine spot test on the treated, washed fabric: when no blue colour develops, no starch remains. The wash water running clear is supporting evidence, but the negative spot test on the cloth itself is the confirmation used before the second weighing.
7. Residual size films block absorbency unevenly, so scouring, bleaching and dyeing act patchily — resist marks, unlevel dyeing, poor print definition and a harsh, boardy handle. Desizing is the first wet-processing step precisely because every later process assumes a size-free, absorbent fabric.

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